

Heaven's Light is Our Guide



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Mechatronics Engineering

**A PROJECT REPORT
ON**

DESIGN AND IMPLEMENTATION OF A 5V TO 24V BOOST CONVERTER USING ESP32 CONTROL

Course No: MTE 3202

Course Title: Power Electronics and Drives Sessional

Submitted To

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Chapter 1

Applying Power Electronics Knowledge

1.1 Introduction

Power electronics is a crucial field in modern electrical and electronic engineering, enabling the efficient conversion, regulation, and control of electrical power. It is widely applied in renewable energy systems, electric vehicles, industrial drives, and consumer electronics. One of the most fundamental areas of power electronics is DC-DC conversion, where voltage levels are adjusted to meet the requirements of different loads or subsystems.

Among DC-DC converter topologies, the boost converter has gained special significance due to its ability to step up a low input voltage to a higher output voltage with high efficiency. This makes it indispensable in battery-powered systems, USB-powered devices, renewable energy integration, and embedded electronics.

In this project, titled “Design and Implementation of a 5V to 24V Boost Converter Using ESP32 Control,” we focus on both theoretical and practical aspects of boost converter design. The project combines classical power electronics knowledge such as converter topologies, switching device selection, inductor and capacitor sizing with modern embedded control using the ESP32 microcontroller. The ESP32 provides programmable Pulse Width Modulation (PWM) signals, which are used to drive the switching MOSFET of the boost converter. This enables precise duty cycle control and creates opportunities for further system enhancements such as closed-loop regulation, monitoring, and digital control strategies.

The development of this converter not only strengthens our understanding of fundamental power electronics concepts but also bridges the gap between theoretical classroom knowledge and real-world implementation.

1.2 Relevance of Power Electronics in the Project

This project is a direct application of power electronics knowledge to solve a practical engineering challenge. A typical USB port or small battery pack supplies 5V DC, which is insufficient to power higher voltage loads such as certain sensors, actuators, communication devices, or industrial modules that require 24V DC. Instead of using bulky linear regulators or multiple batteries, a boost converter provides a compact, efficient, and reliable solution.

Key contributions of power electronics in this project include:

1. **Switching Devices and Control:** The project makes use of a MOSFET switch, controlled by ESP32-generated PWM signals. Understanding the characteristics of switching devices, their conduction and switching losses, and gate driving requirements is a direct application of power electronics knowledge.

2. **Energy Storage Components:** Inductors and capacitors form the backbone of the converter. Their correct sizing, based on ripple current and voltage ripple requirements, highlights the importance of analyzing energy storage and transfer mechanisms.
1. **Conversion Principles:** The boost converter is governed by the fundamental relation:

$$V_{out} = \frac{V_{in}}{1 - D}$$

where D is the duty cycle. This project applies this formula both in theoretical design and experimental validation.

2. **Efficiency Considerations:** Knowledge of power losses—switching losses, conduction losses, diode recovery losses—helps in selecting appropriate devices and operating conditions to maximize efficiency.
3. **Digital Control Integration:** Using an ESP32, which is not traditionally taught in pure power electronics, demonstrates the interdisciplinary nature of modern systems where embedded control and power electronics work hand-in-hand.

1.3 Application of Theoretical Knowledge

The design and implementation of this boost converter is not only a practical exercise but also a direct application of the theoretical concepts learned in the Power Electronics & Drives course. Throughout the project, several classroom principles such as converter operation, duty cycle control, switching device characteristics, and energy storage behavior are translated into real hardware decisions. By bridging these theoretical foundations with experimental validation, the project demonstrates how abstract equations and models are essential for guiding practical design choices and ensuring reliable converter performance.

Several concepts taught in the Power Electronics & Drives course are directly applied in this project:

- **Converter Topology Selection:** Understanding the operation of buck, boost, and buck-boost converters allows us to select the appropriate circuit for stepping up 5V to 24V. The boost topology is most efficient and suitable for this requirement.
- **Duty Cycle and Voltage Gain Relationship:** From the basic boost converter relation, we calculate the required duty cycle to step 5V up to 24V. This demonstrates the critical link between theory and hardware implementation.
- **Inductor Design:** Using equations for inductor ripple current, we determine the minimum inductance required. This ensures continuous conduction mode (CCM) operation and minimizes current ripple.
- **Capacitor Design:** The output capacitor is calculated based on voltage ripple requirements, ensuring a stable DC output.
- **Switching Frequency Trade-offs:** High frequency reduces component size but increases switching losses. Choosing the correct switching frequency (using ESP32 PWM) balances efficiency and component performance.

- **MOSFET and Diode Selection:** Based on expected current levels, switching speed, and voltage rating, appropriate power semiconductor devices are chosen.
- **Simulation and Analysis:** Theoretical design is often validated using software simulation (MATLAB/PSIM/Proteus), followed by real-world testing.

Thus, the project represents a hands-on application of almost every major concept taught in the Power Electronics & Drives course.

1.4 Objectives of the Project

The objectives of this project are both technical and educational in nature:

1. **Design and Analysis:** To design a boost converter circuit that steps up a 5V input voltage to a regulated 24V output.
2. **Microcontroller Integration:** To integrate the ESP32 microcontroller for generating PWM signals to control the MOSFET switch.
3. **Component Selection and Optimization:** To calculate and select suitable values for the inductor, capacitor, diode, and MOSFET for stable and efficient operation.
4. **Prototype Development:** To build and test a working hardware prototype of the converter.
5. **Performance Evaluation:** To compare theoretical results with experimental measurements in terms of output voltage, efficiency, and ripple.
6. **Application of Power Electronics Knowledge:** To demonstrate how classroom knowledge of power electronics and drives can be applied to solve a real-world engineering problem.

1.5 Expected Outcomes

By completing this project, the following outcomes are expected:

- **Functional Prototype:** A working boost converter circuit capable of reliably stepping up 5V to 24V.
- **Theoretical Validation:** Experimental results that match or closely follow theoretical predictions.
- **Efficiency Analysis:** Measurement of conversion efficiency, ripple voltage, and transient response.
- **ESP32-Based Control Demonstration:** Proof that a low-cost microcontroller can effectively control power electronic circuits.
- **Skill Development:** Enhanced skills in circuit design, embedded control programming, PCB implementation, and experimental testing.
- **Practical Understanding:** Deeper appreciation of how theoretical knowledge in power electronics is essential for real-world systems like renewable energy inverters, electric drives, and portable electronics.

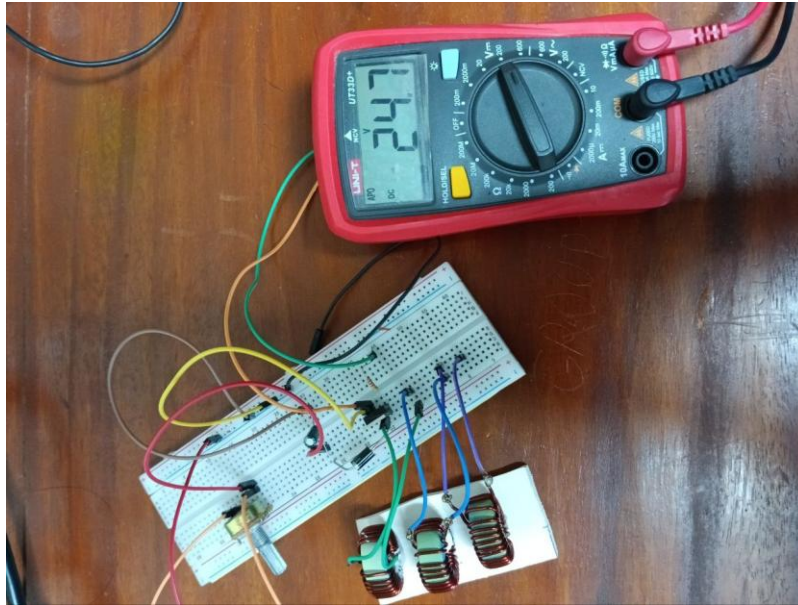


Figure 1.1: Output of the Converter

1.6 Significance of the Project

This project not only demonstrates technical learning but also carries broader practical significance. From an educational perspective, it strengthens the connection between academic knowledge and laboratory practice, allowing theoretical concepts to be validated through hands-on experimentation. Its industrial relevance is evident, as many modules and IoT devices require higher DC voltages than what is typically supplied, making this converter design directly applicable. Finally, it promotes interdisciplinary learning by bridging the principles of power electronics with microcontroller programming and embedded system design.

1. Simplicity and Practical Value:

The boost converter design is based on well-known DC–DC power conversion principles, making it straightforward to implement with basic power electronics knowledge.

2. Common and Affordable Components:

The required components such as Arduino, MOSFET, diode, inductor, and capacitors are inexpensive and widely available in local and online markets. This ensures that the project can be implemented without relying on costly or specialized hardware.

3. Cost-Effectiveness:

Since the project uses low-power components and open-source hardware (Arduino), the overall cost remains within a student-friendly budget. This makes the design suitable for academic, prototype, and laboratory applications.

4. Safety Considerations:

The project operates with a low input voltage of 5V, which is safe for handling during

prototyping. With careful selection of components rated for 24V output, the circuit can be implemented with minimal safety risks as long as standard precautions are followed.

5. Ease of Prototyping and Testing:

The boost converter can be first assembled on a breadboard or PCB for rapid prototyping and easy debugging. Arduino-based control allows for quick code modifications, testing of different duty cycles, and real-time adjustments in the feedback loop.

6. Educational and Research Value:

This project provides hands-on exposure to power electronics, microcontroller-based control, and feedback systems. It serves as a valuable academic exercise and can also form the basis for more advanced research in renewable energy, embedded systems, and efficient power management.

1.7 Summary

In this chapter, we have highlighted how this project applies to fundamental and advanced concepts of power electronics. The design of a 5V to 24V boost converter using an ESP32 microcontroller integrates the principles of DC-DC conversion, duty cycle control, switching device operation, and embedded control. This chapter establishes the theoretical foundation and motivation for the project, which will be further elaborated in subsequent chapters on literature review, methodology, design, implementation, and testing.

Chapter 2

Designing Engineering Solutions

2.1 Introduction

Engineering design is the process of transforming theoretical knowledge into a practical system, which will thinkingly solve a given problem. In this project, the design and implementation of a boost converter to convert 5V DC to a stable output of 2V DC is to be accomplished. While conventional boost converters are used primarily to step up the voltage, our design is geared toward controlled conversion and regulation to provide a lower, precise output voltage to be used for low power electronic applications. The design of this converter involves a delicate consideration of a range of engineering principles, such as power electronics, circuit analysis, and component selection. Dutycycle computation, switching frequency, inductor sizing, and output filtering are optimized to get a better efficient, stable, and safe design. Moreover, theoretical calculations are validated through numerical simulation and prototyping, and the system is optimized for improvements in performance. This chapter explains the systematic design approach used for the boost converter from specification definition, mathematical modeling, through component selection and operational performance trade-offs. Apart from achieving the desired voltage conversion, the goal is for the solution to be cost-effective, reliable and adaptable for real-world applications.

2.2 Objectives

- To solve a complex engineering problem.
- To design a boost converter which will convert 5V DC to 24V DC.
- To analyze output and result.

2.3 Methodology

2.3.1 Theoretical Calculation

- Inductor, capacitor and resistor value was calculated with theoretical formula.
- Clock pulse frequency and duty cycle was calculated to convert 5V DC to 24V DC.

2.3.2 MATLAB Simulation

- The whole circuit was modeled in Simulink with inductor, IGBT module, diode, capacitor and resistor.
- Input was set at 5V DC with dc voltage source and the output was measured by oscilloscope and display.
- The duty cycle was kept 79% and frequency was 40000Hz.

2.3.3 PWM Signal Generation

- ESP 32 was used to generate PWM signal at gate terminal.
- GPIO PIN 5 was used to generate PWM signal.

2.3.4 Hardware Construction

- The whole circuit was build on breadboard so that if need any component can be changed quickly.
- ESP 32 GPIO 5 PIN was connected to TIP 120 transistor which is used to switch the inductor and store energy.
- Diode is connected with inductor and accordingly resistor and capacitors are connected.

2.3.5 Testing

- 5V DC voltage was applied at input side.
- By digital multimeter the output voltage was checked. Output was measured through capacitor terminal
- MATLAB simulation verified the result.

2.4 Component List

Table 2.4.1: Overall Cost of Our Project

Component Name	Functionality	Quantity	Price (BDT)
ESP32 Development Board	Generates PWM control signal	1	449
Inductor (5.44 μ H)	Stores and releases magnetic energy	1	10
Capacitor (39.5 μ F)	Smooths output voltage ripple	1	4
MOSFET IRF530	Main switching device	1	25
Fast Recovery Diode (UF4007)	Provides current path during OFF cycle	1	5
Resistor (10 Ω)	Load/stabilization resistor	1	1
DC Source (9 V battery)	Input supply	1	75
Load Resistor (1 k Ω)	For output voltage testing	1	1
Total Price			= 570 BDT

2.5 Theoretical Calculation

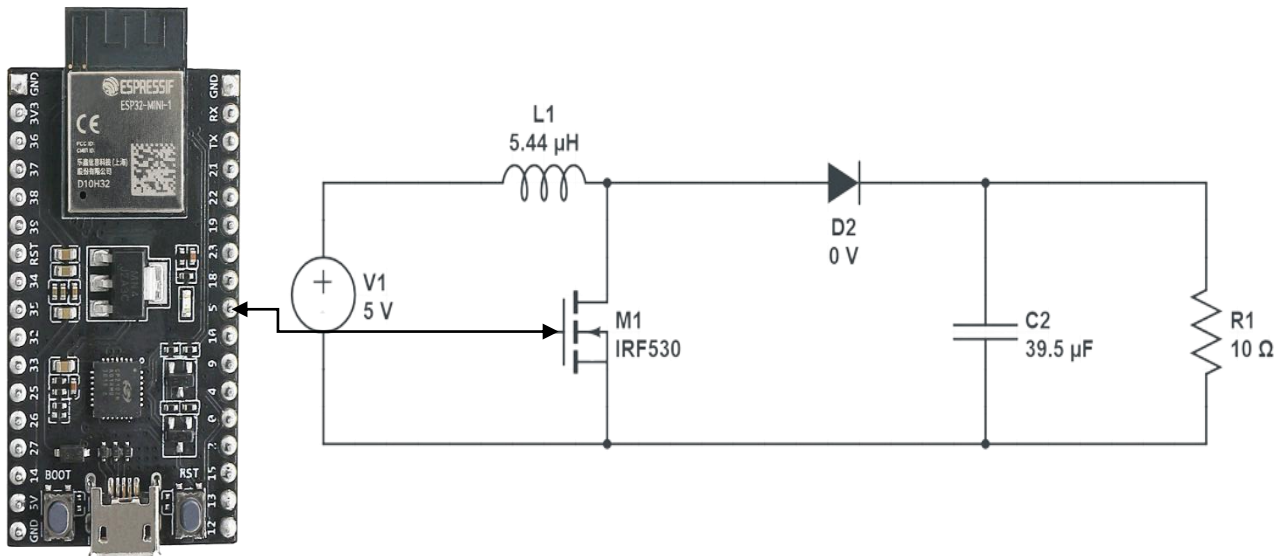


Figure 2.1: Boost Converter circuit diagram with ESP 32

Given Parameters

$$\begin{aligned} V_s &= 5 \text{ V} \\ V_o &= 25 \text{ V} \\ R &= 10 \Omega \\ n &= 0.5\% \\ f &= 40,000 \text{ Hz} \end{aligned}$$

Duty Ratio

$$\begin{aligned} D &= 1 - \frac{V_s}{V_o} \\ D &= 1 - \frac{5}{25} = 0.8 \end{aligned}$$

Minimum Inductance

$$\begin{aligned} L_{min} &= \frac{D(1-D)R}{2f} \\ L_{min} &= \frac{0.8(1-0.8) \times 10}{2 \times 40,000} \\ L_{min} &= 4.35 \mu\text{H} \end{aligned}$$

Maximum Inductance (25% Larger)

$$L_{max} = 1.25 \times L_{min}$$
$$L_{max} = 1.25 \times 4.35 = 5.44 \mu H$$

Minimum Capacitance

$$C = \frac{D}{Rnf}$$
$$C = \frac{0.79}{10 \times 0.05 \times 40,000}$$
$$C = 39.5 \mu F$$

2.6 Design and Simulation

MATLAB Simulink was used to simulate and verify the result. 5V DC was applied to the input terminal. Inductor (5.44μH) was connected in series with power supply. To store power in that a N-channel mosfet was connected. Which gate terminal is connected with ESP 32 GPIO PIN 5. A diode is connected in series and as usual capacitor (39.5μH) is used to smoothen the output and the voltage is measured across load resistor (10Ω). In Simulink a oscilloscope and display is connected at output. And another two output oscilloscope is connected to measure both input and output.

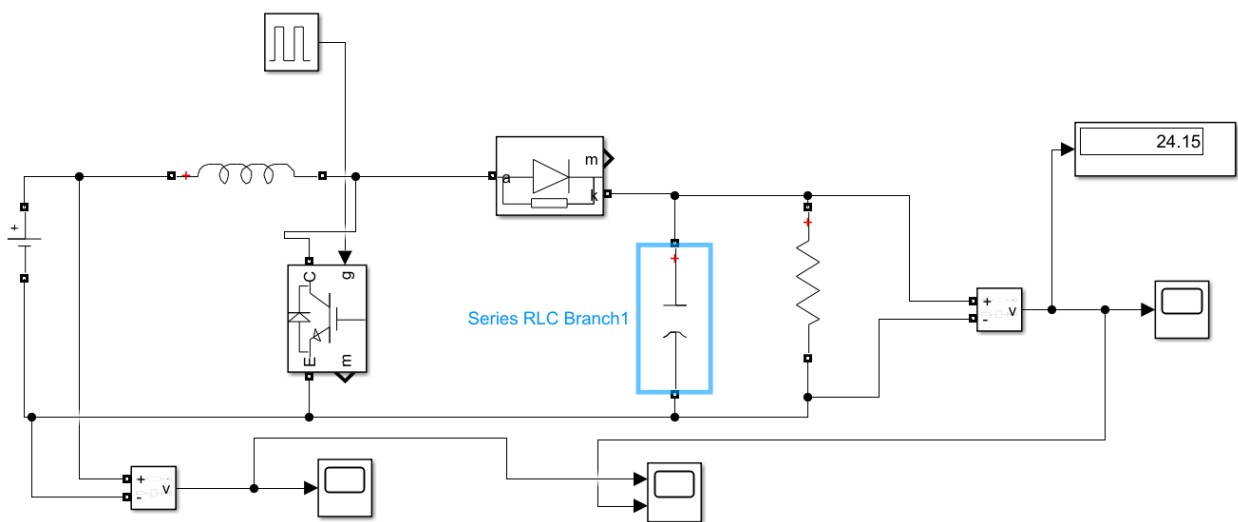


Figure 2.2: Simulation of boost converter in MATLAB Simulink

2.7 ESP 32 PWM Generation

ESP32 produces PWM by using its hardware LEDC (LED Control driver), a counter of cycles and a switch between a HIGH and a LOW state on a GPIO pin per cycle. Each cycle has a duration of 25 μ s and the pin-level is ON with a 79% duty cycle and in each cycle, the pin is ON 19.75 μ s and OFF 5.25 μ s. This hardware-based, high-resolution PWM is necessary to control the boost converter since the duty cycle is what directly controls switching in the MOSFET, energy flow in the inductor, and eventually the output voltage. It is also operated at 40 kHz to minimize ripple and to enable smaller passive components, making the system efficient and more stable.

2.7.1 ESP 32 Code for generating PWM frequency

```
const int pwmFreq = 40000;

const int pwmChannel = 0;

const int pwmResolution = 10;

const int pwmPin = 5;

const int dutyCycle = (int)(0.79 * ((1 << pwmResolution) - 1));

void setup() {
    ledcSetup(pwmChannel, pwmFreq, pwmResolution);
    ledcAttachPin(pwmPin, pwmChannel);
    ledcWrite(pwmChannel, dutyCycle);
}

void loop() {
}
```

2.8 Simulation Result

The designed boost converter was simulated in MATLAB/Simulink and the output waveforms were viewed with an oscilloscope block. The input voltage of 5 V (yellow trace) is effectively stepped up to about 24-25 V at the output (blue trace), as shown in the figure thus confirming the theoretical design calculations. The output voltage shows that there is a small ripple about the steady-state value. This ripple should occur because of charging and discharging nature of the inductor and switching behavior of the diode and MOSFET. The value of ripple is not too large to be acceptable, which means that the values of inductance and capacitance used is sufficiently large to keep the voltage constant within the specified load.

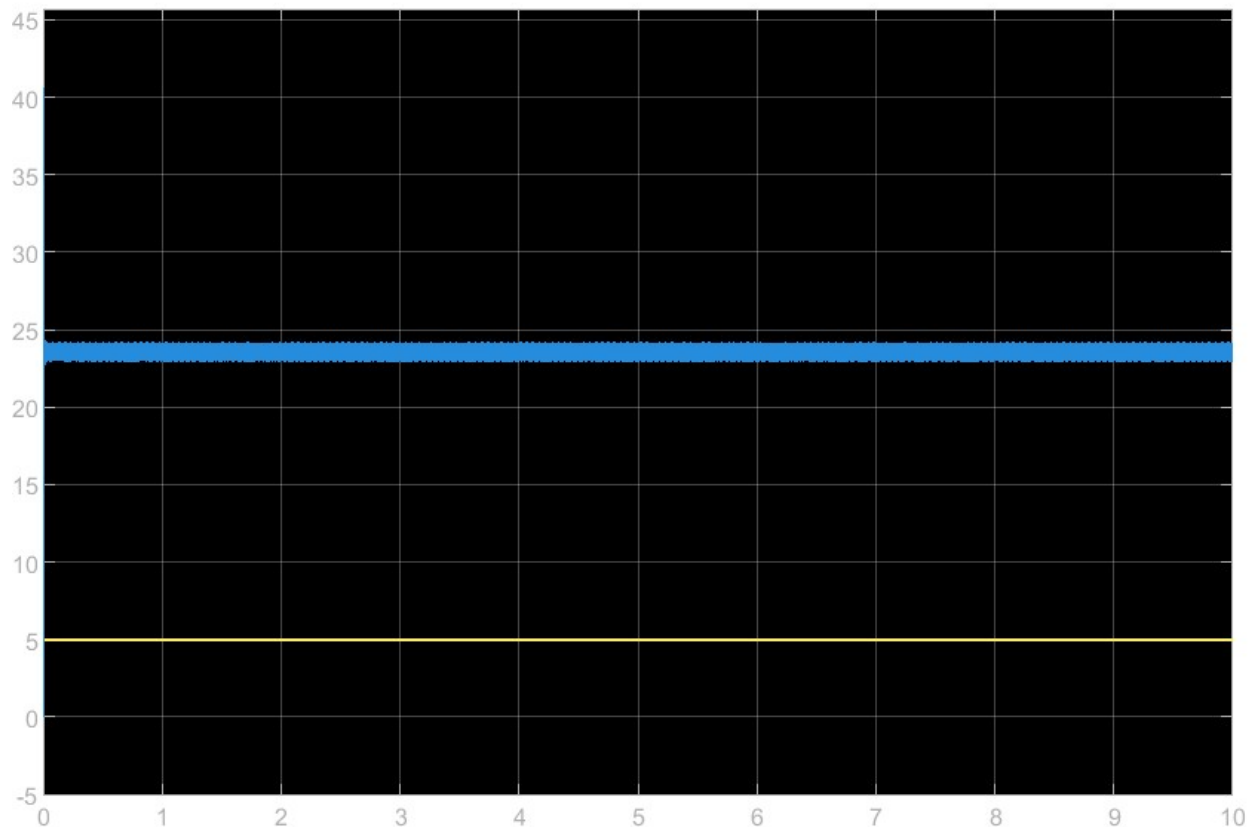


Figure 2.3: MATLAB Simulink Result

The measured performance allows us to conclude that the boost converter functions as intended, and the target voltage gain is obtained using a duty cycle of 79 percent and a switching frequency of 40 kHz. Ripple shows the pragmatic character of switching converters, in which an ideal DC output can not be reached, but can be reduced through smoothing of passive components. Generally, both the theoretical design and the control approach are confirmed in the simulation showing that converter can effectively scale-up a 5 V supply to controlled 25 V output.

2.9 Hardware Implementation

The boost converter was implemented on a breadboard platform in hardware to do quick prototyping and testing. To place the PWM signal at 40 kHz with a duty cycle of about 79 percent, an ESP32 microcontroller was utilized to supply the switching control necessary to the MOSFET. The circuit is powered by inductors over toroidal cores which are the storage of the energy needed to step-up the input voltage. A potentiometer was added to enable the control parameters to be adjusted and a 9 V DC battery was taken as the input supply source. The power stage of the boost converter is interconnected by the wiring with the passive and active components, the inductor, diode, and the capacitor. The ESP32 is supplied through the USB to be stable and generate signals. This system proves the implementation of the converter in a practical manner that can bridge the gap between the simulation results and experimental validation. Despite the use of a breadboard in the first testing, the design has been able to recreate the desired step-up behavior, thus it is possible to note that the proposed engineering solution is viable.

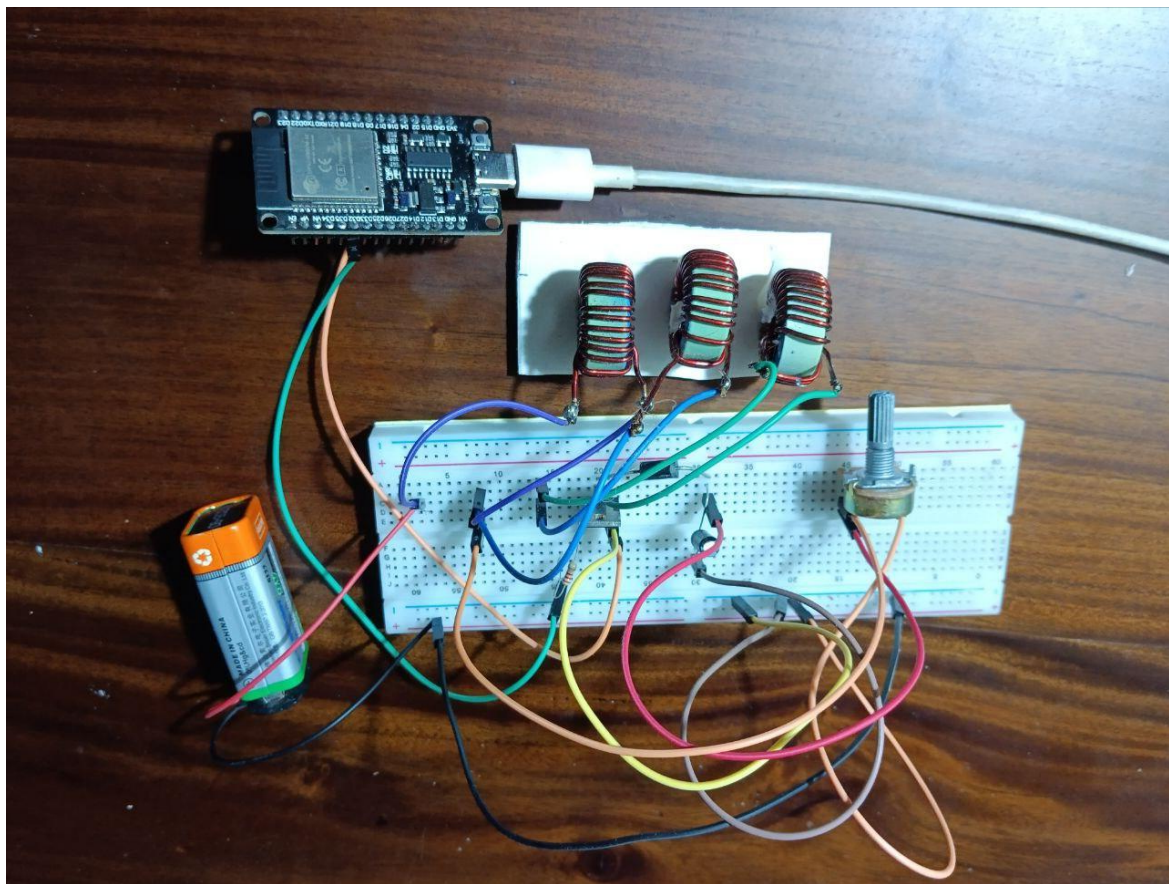


Figure 2.4: Hardware implementation of boost converter

Chapter 3

Contextual Reasoning for Power Electronics

3.1 Component Selection and Trade-Offs for Efficiency

The Boost Converter requires careful selection of components to ensure high efficiency and stable performance. Given the high duty cycle of the converter, components must be chosen to handle significant stress while minimizing losses.

Inductor and Capacitor: The inductor should be large in order for energy storage not to saturate and result in poor performance, and the capacitor smooths out the output voltage. A proper inductor size and capacitance should be chosen not to have much ripple and ensure continuous conduction mode (CCM), where energy transmission is done efficiently and smoothly.

MOSFET and Diode: MOSFET and diode are responsible for switching and energy flow control. In order to reduce conduction loss and switching loss, low resistance MOSFETs and Schottky diodes are chosen. They support the high current and switching rate needed by the high duty cycle while remaining efficient and lessening heat production.

The selection of appropriate components ensures that the Boost Converter operates optimally, with low energy loss and thermal stress.

3.2 Voltage Regulation and Stability

One of the significant challenges for a Boost Converter is stable voltage control. The load and input voltage variations will lead to output voltage variations. A feedback control system is employed for counteracting this. The system varies the duty cycle adaptively for a stable output of 24V, enabling the converter consistently to accommodate variations in the input or in the load conditions.

This real-time correction is essential for the converter output stability and shows the need for feedback control for a successful power electronic design.

3.3 Thermal Management

The high duty cycle also increases the component stress, especially that of MOSFET and inductor, and results in more power dissipation and overheating. Proper thermal management has to be done in order to prevent component failure due to excessive heating.

Heat sinking and efficient design of a PCB are also essential in enabling the components to remain at safe temperatures. Efficient heat dissipation also makes the converter reliable for

long periods of usage, particularly in real-world applications where thermal stress is a serious issue.

3.4 Optimization and Trade-offs

Whereas design at a theoretical level creates a basis, practical applications very often require compromise. The selection of switching frequency, for example, has a direct bearing both on efficiency and component packaging. Higher switching frequencies minimize component packaging at the expense of increased switching loss, and a delicate balance has to be struck for a desired performance.

Likewise, PWM control integration using ESP32 gives highly accurate duty cycle adjustments for optimized performance and ease of adjustments and monitoring. This identifies that there need to be both analog and digital solutions for today's power electronics design applications that need accurate control and efficiency.

3.5 Conclusion

Contextual reasoning for Boost Converter design stresses the importance of applying practical constraints to theoretical knowledge. By employing the appropriate components, real-time control of voltage, thermal control, and optimization of system performance by trade-off, the converter can efficiently and reliably function. The design decisions that are made within this chapter verify that the Boost Converter reliably raises from 5V to 24V and meets real-world applications performance requirements.

CHAPTER 4

Sustainable Applications and Environmental Impact

4.1 Introduction

In the modern world the engineering practice sustainability has become more important as performance and efficiency. In Power Electronics, DC-DC converters as the boost converter, are designed not only for electrical functionality but also for their broader role in environmental and societal development. The project “Design and Implementation of a 5V to 24V Boost Converter Using ESP32” is aligned with these sustainable goals because it demonstrates how small, efficient, and low-cost electronic systems can reduce resource consumption and promote the integration of renewable energy.

4.2 Role of Boost Converter in Sustainable Systems

The boost converter contributes to sustainability in several ways:

- **Efficiency:** Boost converters operate more efficiently than linear regulators by transferring energy rather than wasting excess energy through heat. Since the goal is to avoid energy waste and burn when an energy source is limited, especially, for example, a battery or solar panel.
- **Renewable Energy:** Many renewable energy devices, regardless of their power potential, have low (5-12V) outputs, like most small-scale photovoltaic (PV) panels. A boost converter will step up voltages to viable levels in batteries or devices and make using renewable energy simple and practical.
- **Compact and Low-Cost:** Relying on inexpensive and available components to build this converter, it does not have large, resource-intensive equipment, reducing much resource impact to create a material footprint.

4.3 Environmental Impact of the Project

The ecological importance of this project can be classified as either direct or indirect impacts:

- The direct impact is that the converter operates at low voltage and uses minimum power during the experiments, allowing for safe operation (with little emission and no hazardous byproducts).
- The indirect impact is that the converter provides the technical basis for applications that would reduce fossil fuel reliance; for example, solar-powered applications, energy-efficient portable electronics, and green IoT devices. This indirect use of small-scale renewable energy sources reduces greenhouse gases and encourages greener energy use.

4.4 Sustainable Applications

This project demonstrates several practical applications that align with sustainable development goals:

1. **Solar-Powered Systems:** A 5V solar panel can be boosted to 24V to charge batteries or operate equipment, promoting renewable energy in rural or off-grid areas.
2. **Low-Cost Educational Tools:** The design can be replicated by students using affordable components, making it an accessible learning tool without requiring high resource consumption.
3. **Energy-Conscious Devices:** Many IoT sensors and microcontrollers operate from limited energy sources. Using a boost converter allows them to run longer and more reliably, lowering the frequency of battery replacements and reducing electronic waste.
4. **Portable and Wearable Electronics:** By providing efficient voltage conversion, the project supports lightweight and eco-friendly device design, which uses fewer materials and consumes less power.

4.5 Sustainability Considerations in Design

While developing the converter, several choices were made that reflect sustainable engineering practices:

- **Low Power Input:** Operating with a 5V source ensures safety and minimal energy usage during testing and operation.
- **Reusable Components:** The MOSFET, inductor, and capacitors are standard parts that can be reused in future projects, avoiding waste.
- **Scalability:** The same design principles can be applied to larger systems such as solar inverters or electric drives, amplifying their long-term sustainable impact.
- **Minimizing Heat Dissipation:** Efficient switching operations reduce unnecessary heating, which not only saves energy but also prolongs the lifespan of components, thereby reducing electronic waste.

4.6 Educational and Environmental Value

The project is not limited to academic purposes. It has broader educational and environmental value:

1. Students and researchers learn about energy efficient systems design with sustainability in mind.
2. The converter provides an entry point for real world projects based on renewable energy, green electronics, and environmentally friendly technology.
3. Raising awareness of efficient energy conversion is part of the culture of responsible engineering, in which the care for the environment prevails.

4.7 Summary

In summary, the boost converter project is not merely a technical act of solving a problem but begins to cross into the realm of sustainable engineering. So, energy efficiency, renewable energy, size reduction, and waste considerations can contribute to the global movement towards reducing impact on the environment. Its applications to solar power, portable devices, and education are more relevant to building a more sustainable, green future.

Chapter 5

Effective Individual and Team Performance

5.1 Introduction

Effective engineering projects rely on coordinated teamwork. It is common practice for complex systems to be developed by groups rather than individuals. In our 5V→24V boost-converter project, the team explicitly divided the work so that all key tasks were covered. For example, some members focused on circuit design and simulation (selecting the inductor, diode, MOSFET, and capacitors), while others developed and tested the Arduino firmware for PWM control and feedback. Additional members built the hardware prototype on a breadboard and performed measurements, and others compiled reports and documentation. This distribution follows engineering education best practice: as one source notes, students “are increasingly expected to work in teams and participate in projects”. Frequent meetings and shared documentation ensured that, despite working on different parts, everyone stayed aligned toward the project’s goal.

5.2 Division of Responsibilities

To manage our distributed effort, the team allocated responsibilities along functional lines.

- i. **Circuit design and simulation:** Calculating component values and modelling the boost converter’s power stage (inductor, diode, MOSFET, output capacitor) to predict its behaviour under various loads.
- ii. **Control software:** Writing and testing the Arduino code, including configuring PWM generation and implementing the feedback control algorithm.
- iii. **Hardware prototyping:** Assembling the boost circuit on a breadboard or PCB, wiring the MOSFET and passive components, and setting up the Arduino connections.
- iv. **Testing and debugging:** Measuring voltages and waveforms (using multimeters and oscilloscopes), identifying issues in the converter’s operation, and iterating on design.
- v. **Documentation and coordination:** Recording test results, creating schematics, and writing the report, while ensuring that all team members were up to date on progress and changes.

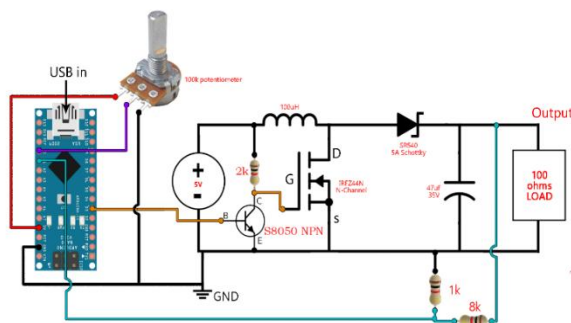


Figure 5.1: Circuit diagram of Boost Converter Using Arduino

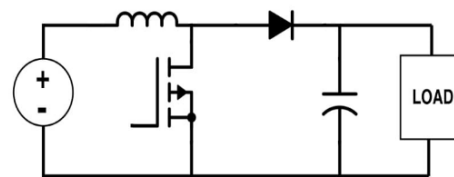


Figure 5.2: Generic topology for a boost convert

The Arduino generates the PWM drive to the MOSFET and reads the converter's output via an analog input. The physical prototype was assembled on a standard solderless breadboard, as shown above. In this setup, the Arduino board provided the PWM signal to the MOSFET gate and measured the converter's output through an analog input. The required inductor, diode and a large output capacitor are visible, illustrating how the power stage and controller were connected. This hands-on arrangement enabled rapid testing: when one member tweaked a component or uploaded new code, the effects were immediately observable. Dividing the workload in this way allowed the team to make parallel progress and later integrate subsystems smoothly. By clearly defining roles but maintaining open communication, each member's efforts contributed to the shared objectives without duplication of work.

5.3 Technical Challenge

5.3.1 Achieving the Voltage Boost

During testing, the team encountered a significant technical hurdle: the converter initially failed to reach the intended 24V output. Under open-loop (fixed duty-cycle) conditions, the output stalled well below the target. Recognising that a simple PWM setting was insufficient, the team embarked on systematic troubleshooting. One contributor had warned that “control loops are hard” and that factors like sampling rate and PWM frequency make boost regulator design “a highly mathematical exercise”. With that in mind, we first verified each subsystem carefully. We checked the MOSFET's switching waveform on an oscilloscope, confirmed that the inductor and diode were the correct type and rating, and ensured proper grounding and wiring. To improve hardware stability, the team added a large electrolytic capacitor ($\approx 2200\text{ }\mu\text{F}$ at 25V) at the converter output. This smoothing capacitor helps hold the boosted voltage during switching and is a standard practice for stabilising boost converters.

Our debugging process included:

- **Subsystem testing:** Verifying individual components and signals. We measured the raw output voltage and the MOSFET gate waveform, checking that the circuit topology matched the design. Any wiring errors or faulty components were ruled out.
- **Hardware improvements:** Adding or adjusting components to support boosting. As noted, a large output capacitor was installed. We also ensured the input supply was stiff and added small decoupling capacitors to the Arduino supply to prevent resets.
- **Feedback control implementation:** Implementing closed-loop control in the Arduino code. The converter's output (scaled into the Arduino's 0-5 V input range) was fed into an analog pin, and the software adjusted the PWM duty cycle based on the voltage error. In this way, the converter operated as a closed-loop system. This approach matches standard voltage-mode control schemes for boost converters. We coded a simple proportional controller that increased the duty cycle when the measured output was below the 24V setpoint.
- **Iterative tuning:** Gradually adjusting parameters based on test results. After enabling feedback, we performed a sequence of tests: if the output overshoot or oscillated, we reduced the control gain or added a small dead-time; if the output was slow to rise, we

incrementally increased the gain. Each iteration involved measuring the response and discussing the results as a group. Eventually, the output converged to 24V reliably, even under varying input or load conditions.

Through this structured approach combining hardware fixes with control software improvements, the team overcame the initial limitation. The converter ultimately achieved the required boost as specified. The process demonstrated how collaborative troubleshooting and an improved control strategy can resolve complex power-electronics challenges.

5.3.2 Communication and Adaptive Learning

Throughout the project, continuous communication and learning were crucial. The team held regular meetings to share findings and coordinate next steps. When a problem arose in one area, that insight was immediately communicated: for instance, if the circuit team observed unexpected voltage ripple, they discussed it with the software team to see if the code should compensate, and vice versa. This exchange ensured that knowledge was pooled, not siloed. Team members also did not hesitate to learn from external sources or from one another. For example, when one person researched Arduino timer settings to increase PWM frequency, they taught the team how to modify the code. This collaborative, open culture reflects recognised teamwork principles. As one educational framework observes, strong engineering teams “collaborate across disciplines and show respect for the contributions of all members and their unique roles”. In our group, everyone’s perspective was valued. If the coder suggested a new algorithm, the circuit designer was open to test it; if a builder noticed a wiring issue, the software developer adjusted the code accordingly. By listening to each other and adapting together, the team quickly bridged knowledge gaps. This adaptive learning meant that hardware and control limitations were progressively overcome by collective effort rather than individual trial-and-error.

5.3.3 Aspects of our individual and team performance included

- **Clear role division:** Each member focused on a specific area (design, coding, testing, documentation) while remaining aware of the overall goal.
- **Iterative troubleshooting:** Problems were resolved through systematic testing, combining hardware modifications (such as adding output capacitance) with refined control software.
- **Effective communication:** Regular coordination ensured that new ideas or problems were shared promptly; peers reviewed and cross-checked each other’s work to catch issues early.
- **Adaptive learning:** Team members quickly researched solutions or taught each other unfamiliar concepts, turning setbacks into learning opportunities.
- **Unified objective:** Despite varied tasks, everyone remained committed to the project’s single goal (achieving 24V output). This focus meant that all individual contributions aligned toward a successful, integrated solution.

By distributing responsibilities, communicating openly, and learning from each challenge, the team’s individual efforts coalesced into a coherent system. This synergy, each person’s work reinforcing the others’, was instrumental in achieving the boost converter’s design objectives.

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